Mass, according to Einstein's special relativity laws, is just a very compact form of energy. It is possible, though *how* is a nontrivial issue, to convert any mass, including that of a person, into explosive energy. The amount of energy that comes from such conversion is enormous. It is given by Einstein's famous formula $E = Mc^2$, where E is the explosive energy, M is the mass that gets converted to energy, and $c = 2.99792 \times 10^8$ meters per second is the speed of light. From the 75-kilogram mass of a typical person this formula predicts an explosive energy of 7×10^{18} joules, which is thirty times larger than the energy of the most powerful hydrogen bomb that has ever been exploded.

The conversion of mass into heat or into the kinetic energy of an explosion underlies Zwicky's explanation for supernovae (Figure 5.1), the nuclear burning that keeps the Sun hot (later in this chapter), and nuclear explosions (next chapter).

Zwicky was not content with just explaining supernovae; he wanted to explain everything in the Universe. Among all the unexplained things, the one getting the most attention at Caltech in 1932–1933 was *cosmic* rays—high-speed particles that bombard the Earth from space. Caltech's Robert Millikan was the world leader in the study of cosmic rays and had given them their name, and Caltech's Carl Anderson had discovered that some of the cosmic-ray particles were made of *antimatter*.[2] Zwicky, with his love of extremes, managed to convince himself (correctly it turns out) that most of the cosmic rays were coming from outside our solar system, and (incorrectly) that most were from outside our Milky Way galaxy—indeed, from the most distant reaches of the Universe—and he then convinced himself (roughly correctly) that the total energy carried by all the Universe's cosmic rays was about the same as the total energy released by supernovae throughout the Universe. The conclusion was obvious to Zwicky (and perhaps correct[3]): Cosmic rays are made in supernova explosions.

It was late 1933 by the time Zwicky had convinced himself of these connections between supernovae, neutrons, and cosmic rays. Since Baade's encyclopedic knowledge of observational astronomy had been a crucial foundation for these connections, and since many of Zwicky's calculations